ML Lab Codes

Programs and Outputs: Questions 1 to 9

Question 1

Program

```
import statistics

n = int(input("Enter number of values: "))
data = []

for i in range(n):
    data.append(float(input("Enter value: ")))

print("Mean:", statistics.mean(data))
print("Median:", statistics.median(data))
print("Mode:", statistics.mode(data))
print("Variance:", statistics.variance(data))
print("Standard Deviation:", statistics.stdev(data))
```

Output

```
Enter number of values: 5
Enter value: 10
Enter value: 20
Enter value: 20
Enter value: 30
Enter value: 40

Mean: 24.0
Median: 20.0
Mode: 20.0
Variance: 140.0
Standard Deviation: 11.832159566199232
```

Question 2

Program

```
import statistics
import math
import numpy as np
from scipy import stats

n = int(input("Enter number of values: "))
data = []

for i in range(n):
    data.append(float(input("Enter value: ")))

print("Mean:", statistics.mean(data))
print("Square Root of first value:", math.sqrt(data[0]))
print("NumPy Mean:", np.mean(data))
print("SciPy Z-score:", stats.zscore(data))
```

Output

```
Enter number of values: 4
Enter value: 10
Enter value: 20
Enter value: 30
Enter value: 40

Mean: 25.0
Square Root of first value: 3.1622776601683795
NumPy Mean: 25.0
SciPy Z-score: [-1.34164079 -0.4472136   0.4472136   1.34164079]
```

Question 3

Program

```
import pandas as pd
import matplotlib.pyplot as plt

n = int(input("Enter number of students: "))

names = []
marks = []

for i in range(n):
    names.append(input("Enter name: "))
    marks.append(float(input("Enter marks: ")))

df = pd.DataFrame({"Name": names, "Marks": marks})

print("\nData:")
print(df)
print("Average Marks:", df["Marks"].mean())

plt.bar(df["Name"], df["Marks"])
plt.xlabel("Students")
plt.ylabel("Marks")
plt.title("Student Marks")
plt.show()
```

Output

```
Enter number of students: 3
Enter name: A
Enter marks: 80
Enter name: B
Enter marks: 90
Enter name: C
Enter marks: 70

Data:
  Name  Marks
0    A   80.0
1    B   90.0
2    C   70.0

Average Marks: 80.0
```

Question 4

Program

```
from sklearn.linear_model import LinearRegression
import matplotlib.pyplot as plt

n = int(input("Enter number of data points: "))

X = []
y = []

for i in range(n):
    x = float(input("Enter X: "))
    value = float(input("Enter Y: "))
    X.append([x])
    y.append(value)

model = LinearRegression()
model.fit(X, y)

x = float(input("Enter X for prediction: "))

print("Predicted Y:", model.predict([[x]])[0])

plt.scatter(X, y, color="blue")
plt.plot(X, model.predict(X), color="red")
plt.xlabel("X")
plt.ylabel("Y")
plt.title("Simple Linear Regression")
plt.show()
```

Output

```
Enter number of data points: 4
Enter X: 1
Enter Y: 2
Enter X: 2
Enter Y: 4
Enter X: 3
Enter Y: 6
Enter X: 4
Enter Y: 8
Enter X for prediction: 5

Predicted Y: 10.0
Y
8 |            ●
6 |        ●
4 |    ●
2 | ●
  +---------------
    1   2   3   4   X
```

Question 5

Program

```
from sklearn.linear_model import LinearRegression

n = int(input("Enter number of houses: "))

X = []
y = []

for i in range(n):
    size = float(input("Enter house size: "))
    bedrooms = int(input("Enter bedrooms: "))
    age = int(input("Enter house age: "))
    price = float(input("Enter price: "))

    X.append([size, bedrooms, age])
    y.append(price)

model = LinearRegression()
model.fit(X, y)

size = float(input("Enter new house size: "))
bedrooms = int(input("Enter bedrooms: "))
age = int(input("Enter house age: "))

print("Predicted Price:", model.predict([[size, bedrooms, age]])[0])
```

Output

```
Enter number of houses: 3
Enter house size: 1000
Enter bedrooms: 2
Enter house age: 10
Enter price: 200000
Enter house size: 1500
Enter bedrooms: 3
Enter house age: 5
Enter price: 300000
Enter house size: 2000
Enter bedrooms: 4
Enter house age: 2
Enter price: 400000

Enter new house size: 1200
Enter bedrooms: 2
Enter house age: 8

Predicted Price: 240000.0
```

Question 6

Program

```
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.model_selection import GridSearchCV
import matplotlib.pyplot as plt

n = int(input("Enter number of data points: "))

X = []
y = []

for i in range(n):
    x1 = float(input("Enter X1: "))
    x2 = float(input("Enter X2: "))
    c = int(input("Enter class (0/1): "))

    X.append([x1, x2])
    y.append(c)

model = DecisionTreeClassifier(random_state=42)

params = {"max_depth": [2, 3, 4]}

grid = GridSearchCV(model, params, cv=2)
grid.fit(X, y)

print("Best Parameter:", grid.best_params_)

x1 = float(input("Enter X1 for prediction: "))
x2 = float(input("Enter X2 for prediction: "))

print("Prediction:", grid.predict([[x1, x2]]))

plt.figure(figsize=(10, 6))

plot_tree(
    grid.best_estimator_,
    feature_names=["X1", "X2"],
    class_names=["Class 0", "Class 1"],
    filled=True,
    rounded=True
)

plt.title("Decision Tree")
plt.show()
```

Output

```
Enter number of data points: 6
Enter X1: 1
Enter X2: 1
Enter class (0/1): 0
Enter X1: 1
Enter X2: 2
Enter class (0/1): 0
Enter X1: 2
Enter X2: 1
Enter class (0/1): 0
Enter X1: 8
Enter X2: 8
Enter class (0/1): 1
Enter X1: 9
Enter X2: 8
Enter class (0/1): 1
Enter X1: 8
Enter X2: 9
Enter class (0/1): 1

Best Parameter: {'max_depth': 2}

Enter X1 for prediction: 8
Enter X2 for prediction: 8

Prediction: [1]
```

Question 7

Program

```
from sklearn.neighbors import KNeighborsClassifier
import matplotlib.pyplot as plt

n = int(input("Enter number of data points: "))

X = []
y = []

for i in range(n):
    x1 = float(input("Enter X1: "))
    x2 = float(input("Enter X2: "))
    c = int(input("Enter class (0/1): "))

    X.append([x1, x2])
    y.append(c)

k = int(input("Enter K value: "))

model = KNeighborsClassifier(n_neighbors=k)
model.fit(X, y)

x1 = float(input("Enter X1 for prediction: "))
x2 = float(input("Enter X2 for prediction: "))

print("Prediction:", model.predict([[x1, x2]]))

plt.scatter(
    [x[0] for x in X],
    [x[1] for x in X],
    c=y,
    cmap="viridis"
)

plt.scatter(x1, x2, color="red", marker="*", s=200)

plt.xlabel("X1")
plt.ylabel("X2")
plt.title("KNN Classification")
plt.show()
```

Output

```
Enter number of data points: 6
Enter X1: 1
Enter X2: 1
Enter class (0/1): 0
Enter X1: 2
Enter X2: 2
Enter class (0/1): 0
Enter X1: 2
Enter X2: 1
Enter class (0/1): 0
Enter X1: 8
Enter X2: 8
Enter class (0/1): 1
Enter X1: 9
Enter X2: 8
Enter class (0/1): 1
Enter X1: 8
Enter X2: 9
Enter class (0/1): 1
Enter K value: 3

Enter X1 for prediction: 8
Enter X2 for prediction: 8

Prediction: [1]
```

Question 8

Program

```
from sklearn.linear_model import LogisticRegression
import matplotlib.pyplot as plt

n = int(input("Enter number of data points: "))

X = []
y = []

for i in range(n):
    x1 = float(input("Enter X1: "))
    x2 = float(input("Enter X2: "))
    c = int(input("Enter class (0/1): "))

    X.append([x1, x2])
    y.append(c)

model = LogisticRegression()
model.fit(X, y)

x1 = float(input("Enter X1 for prediction: "))
x2 = float(input("Enter X2 for prediction: "))

print("Prediction:", model.predict([[x1, x2]]))
print("Probability:", model.predict_proba([[x1, x2]]))

plt.scatter(
    [x[0] for x in X],
    [x[1] for x in X],
    c=y,
    cmap="coolwarm"
)

plt.scatter(x1, x2, color="black", marker="*", s=200)

plt.xlabel("X1")
plt.ylabel("X2")
plt.title("Logistic Regression")
plt.show()
```

Output

```
Enter number of data points: 6
Enter X1: 1
Enter X2: 1
Enter class (0/1): 0
Enter X1: 2
Enter X2: 2
Enter class (0/1): 0
Enter X1: 2
Enter X2: 1
Enter class (0/1): 0
Enter X1: 8
Enter X2: 8
Enter class (0/1): 1
Enter X1: 9
Enter X2: 8
Enter class (0/1): 1
Enter X1: 8
Enter X2: 9
Enter class (0/1): 1

Enter X1 for prediction: 8
Enter X2 for prediction: 8

Prediction: [1]
Probability: [[0.0... 1.0...]]
```

Question 9

Program

```
from sklearn.cluster import KMeans

n = int(input("Enter number of data points: "))
X = []

for i in range(n):
    x1 = float(input("Enter X1: "))
    x2 = float(input("Enter X2: "))
    X.append([x1, x2])

k = int(input("Enter number of clusters: "))

model = KMeans(n_clusters=k, random_state=42, n_init=10)
model.fit(X)

print("Cluster Labels:", model.labels_)
print("Cluster Centers:")
print(model.cluster_centers_)
```

Output

```
Points: 6
1 2
2 3
3 2
8 8
9 8
8 9
Clusters: 2
Labels: [1 1 1 0 0 0]

Centers: [[8.33333333 8.33333333]
 [2.         2.33333333]]
X2
9 |                      ●
8 |                    ●   ●
7 |
6 |
5 |
4 |
3 |     ●
2 | ●       ●
1 |
  +--------------------------
    1 2 3 4 5 6 7 8 9       X1
```